

C-SPINE Revisited: A Rigorous Replication, Audit, and Extension of Sparse Interpretable Sentence Embeddings

Author Contribution Statement

All authors jointly designed the study, discussed intermediate results, and reviewed the final manuscript. Individual contributions, reflecting primary ownership of specific components, are detailed below in CRediT-style form.

Samir Abdullazade

Data Pipeline & Configuration

- Designed and implemented the dataset registry and stratified train/validation/test splitting logic (*src/data.py*).
- Implemented the frozen-encoder embedding extraction pipeline (*src/embeddings.py*).
- Authored the base experiment configuration objects and the initial JSON configs for the SST-2 and AG News runs (*src/config.py*, *configs/base_sst2.json*, *configs/base_agnews.json*).
- Opened the project's first feature pull request (*feature/sparse-embeddings-setup*), establishing the data and configuration layer that all downstream modules built on.

Murad Asgarov

Model Architecture, Training & Repository Ownership

- Implemented the C-SPINE model family, including the CappedReLU activation, TopK/BatchTopK/DenseAE/MLP variants, and the model factory (*src/sparse_model.py*).
- Implemented the reconstruction and sparsity loss functions, including the ablation-friendly loss with independent ASL/PSL/L1 toggles (*src/losses.py*).
- Authored the training loop and checkpointing logic (*src/training.py*).
- Owned repository setup and integration: initial commit, LICENSE, and review/merge of both feature branches into *main*.

Davud Gurbanov

Interpretability Analysis & Manuscript Production

- Designed and implemented the feature-level interpretability suite: purity, specialization, co-activation, and neighborhood-overlap analysis (*src/interpretability.py*).
- Authored the project presentation (*C_SPINE_Revisited_Presentation.pdf*).
- Produced and merged the final written report (*report/C_SPINE_Report.pdf*), integrating results from all other components into the manuscript.
- Reviewed and merged the *feature_analysis* branch (interpretability, presentation, and report) into *main*.

Gurban Burjuzada

Ablations, Evaluation & Experiment Orchestration

- Implemented the controlled ablation suite: sparsity sweeps, mechanism ablations (TopK/BatchTopK/Dense/L1), and loss-term toggles (*src/ablations.py*).
- Implemented the full evaluation module covering reconstruction metrics, downstream probing, and PCA/random-projection baselines (*src/evaluation.py*).

- Authored the experiment orchestration layer tying configs, training, and evaluation into repeatable runs (*src/experiments.py*).

Raul Aghayev

Cross-Domain Generalization, Visualization & Testing

- Implemented the cross-domain and cross-encoder generalization module, evaluating frozen dictionaries across dataset pairs and encoder families (*src/generalization.py*).
- Implemented the full visualization suite: training curves, activation histograms, class heatmaps, sparsity-tradeoff plots, and transfer matrices (*src/visualization.py*).
- Authored the initial smoke-test module for local, low-compute verification (*tests/test_smoke.py*).

Summary Table

Author	Primary Area	Key Files
Samir Abdullazade	Data pipeline & configuration	data.py, embeddings.py, config.py, base configs
Murad Asgarov	Model architecture, training, repo ownership	sparse_model.py, losses.py, training.py
Davud Gurbanov	Interpretability & manuscript	interpretability.py, presentation, report
Gurban Burjuzada	Ablations, evaluation, orchestration	ablations.py, evaluation.py, experiments.py
Raul Aghayev	Generalization, visualization, testing	generalization.py, visualization.py, test_smoke.py

Note: contributions are attributed by primary ownership of repository components as reflected in commit and pull-request history at the time of writing. All authors participated in study design and result discussion; file-level ownership reflects implementation effort, not sole intellectual contribution.